Hypothesis Testing

CSE 437



Hypothesis

A hypothesis is an **educated guess** about something in the world around you.

- Testable - experiment or observation
- A new medicine you think might work.
- A new teaching method you think might be better.
- It can be anything as long as you can put it to the test.

Hypothesis Statement

If I do this to an independent variable, then this will happen to a dependent variable.

If I give patients counseling in addition to medication, then their overall depression scale will decrease.

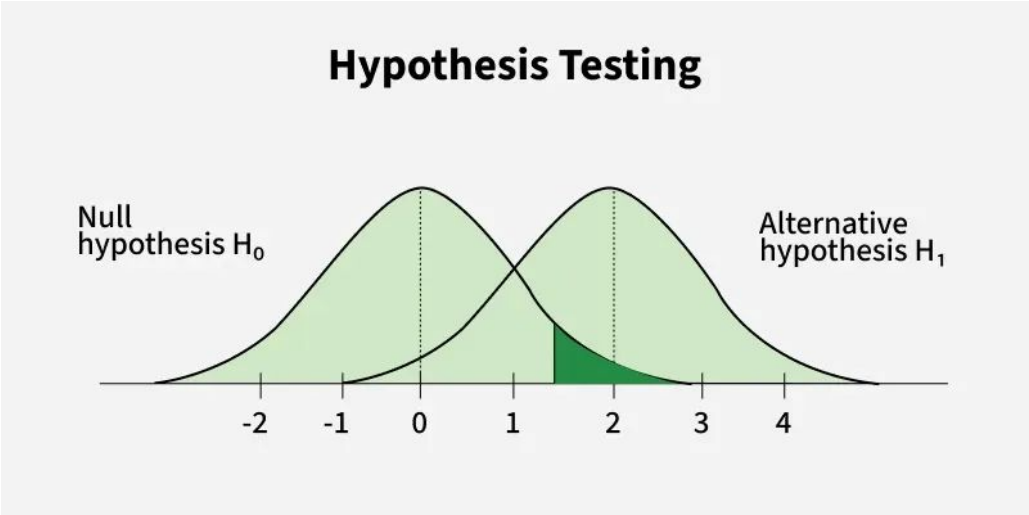
If I give exams at noon instead of 7 pm, then students test score will improve.

If I look in this certain location, then I am more likely to find new species.

Hypothesis Testing

A formal procedure of testing claims and assumptions about data.

- Null hypothesis, H_0 : the default assumption, such as, DNA is shaped like a double helix.
- Alternate hypothesis, H_1: the claim you aim to prove, such as, model A performs better than model B.



Why is it important?

- Is a way to test the results of a survey or experiment to see if we have meaningful result.
- Testing whether the results are valid by figuring out the odds that your results happened by chance

Formulating Null Hypothesis

- Null hypothesis is always the accepted fact.
- Either support or reject null hypothesis.
- Null hypothesis can be nullified (you can reject).

Z-test / Z-score:

$$Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

Where:

- $\bar{x}$ is the sample mean
- μ_0 is the hypothesized population mean
- σ is the population standard deviation
- n is the sample size

Finding Null Hypothesis

A researcher thinks that if knee surgery patients go to physical therapy twice a week instead of three times, their recovery period will be longer. Average recovery period for knee surgery patient is 8.2 weeks.

Find out the null hypothesis.

$H_0 = \mu \leq 8.2$

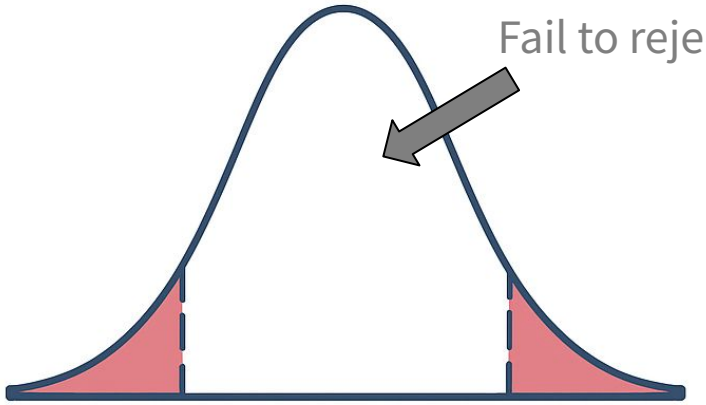
Finding Alternate Hypothesis

Whatever the researcher wanted to say or prove.

$H_1 = \mu > 8.2$ (alternate hypothesis)

Testing a Hypothesis

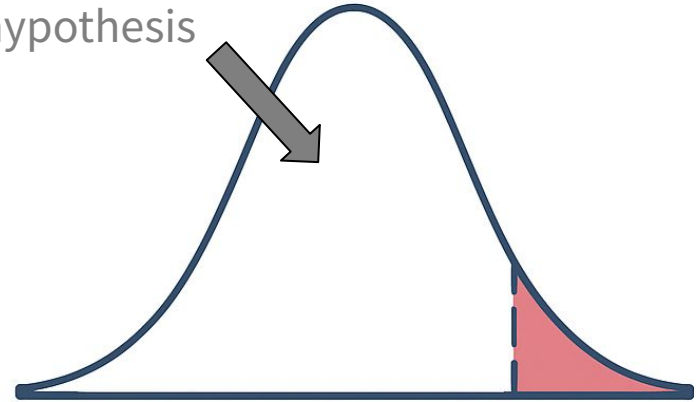
Two-tailed test



Neutral alternative hypothesis

Fail to reject the null hypothesis

One-tailed test

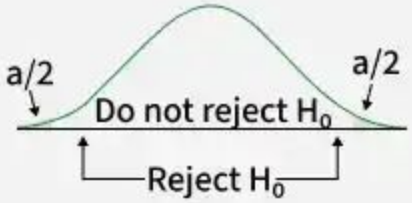


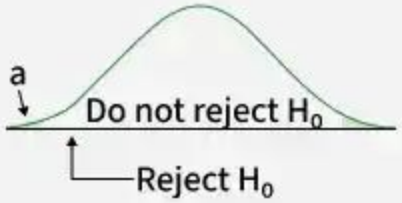
Specific alternative claim

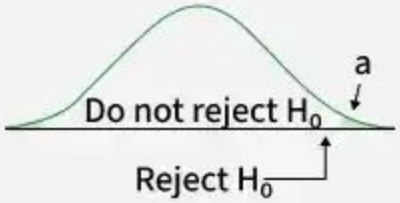
Hypothesis Testing

One-tailed

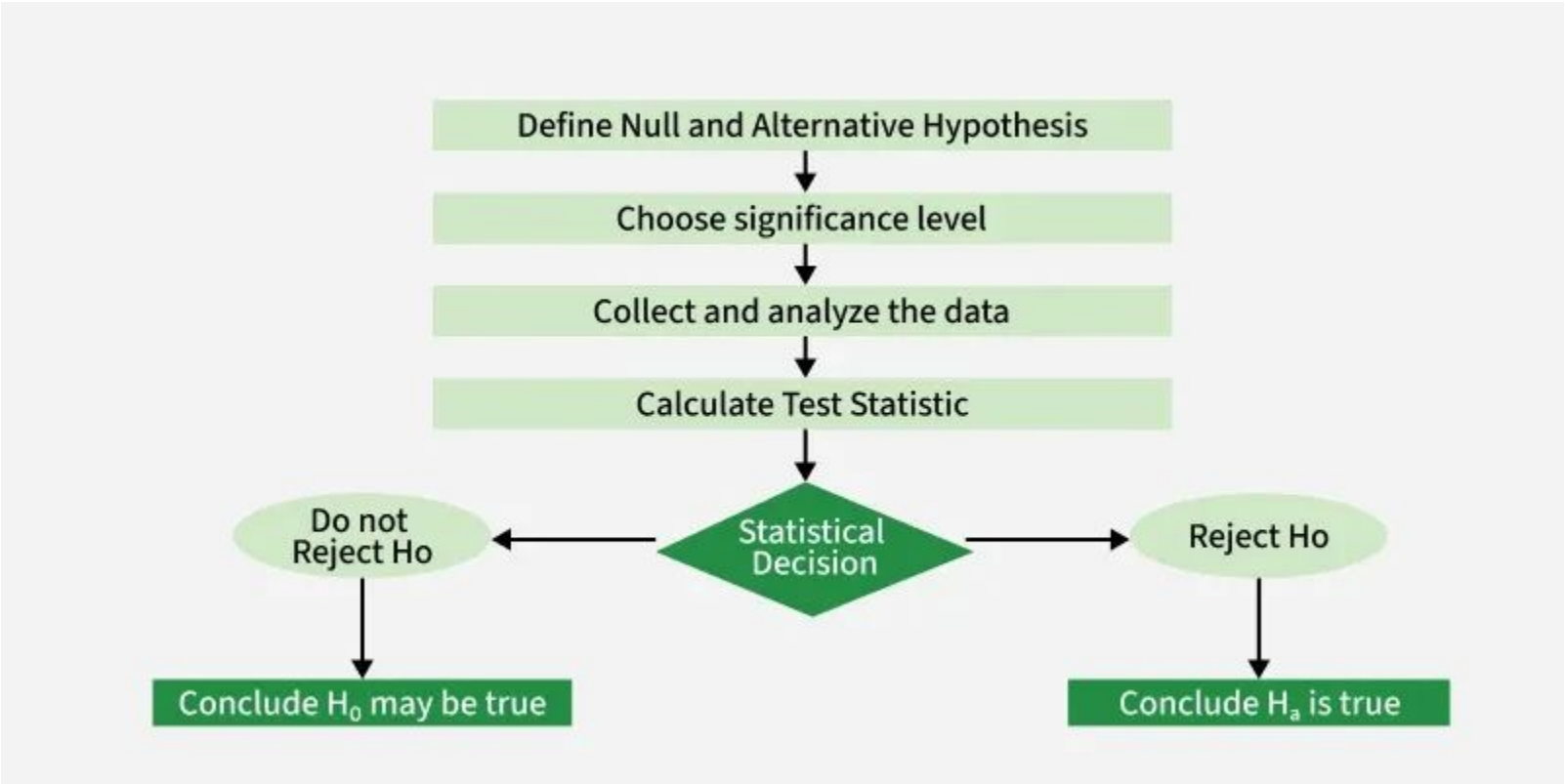
Two-tailed	Left-tailed	Right-tailed
$H_0 : \mu = 50$	$H_0 : \mu \geq 50$	$H_0 : \mu \leq 50$
$H_1 : \mu \neq 50$	$H_1 : \mu < 50$	$H_1 : \mu > 50$







How Does Hypothesis Testing Work?



Step 1: Define Hypotheses:

- **Null hypothesis (H_0):** Assumes no effect or difference.
- **Alternative hypothesis (H_1):** Assumes there is an effect or difference.

Example: Test if a new algorithm improves user engagement.

> ***Note**: In this we assume that our data is **normally distributed**.*

Step 2: Choose significance level

We select a significance level (usually 0.05). This is the maximum chance we accept of wrongly rejecting the null hypothesis (Type I error). It also sets the confidence needed to accept results.

Step 3: Collect and Analyze data.

- Now we gather data this could come from user observations or an experiment. Once collected we analyze the data using appropriate statistical methods to calculate the test statistic.
- **Example**: We collect data on user engagement before and after implementing the algorithm. We can also find the mean engagement scores for each group.

Step 4: Calculate Test Statistic

The test statistic measures how much the sample data deviates from what we did expect if the null hypothesis were true. Different tests use different statistics:

- **Z-test:** Used when population variance is known and sample size is large.
- **T-test:** Used when sample size is small or population variance unknown.
- **Chi-square test:** Used for categorical data to compare observed vs. expected counts.

Z-test

A Z-test is a type of hypothesis test that compares the sample's average to the population's average and calculates the Z-score and tells us how much the sample average is different from the population average by looking at how much the data normally varies.

It is particularly useful when the sample size is large >30. It is also known as Z-Statistics and its formula is:

$$\text{Z-Score} = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

where:

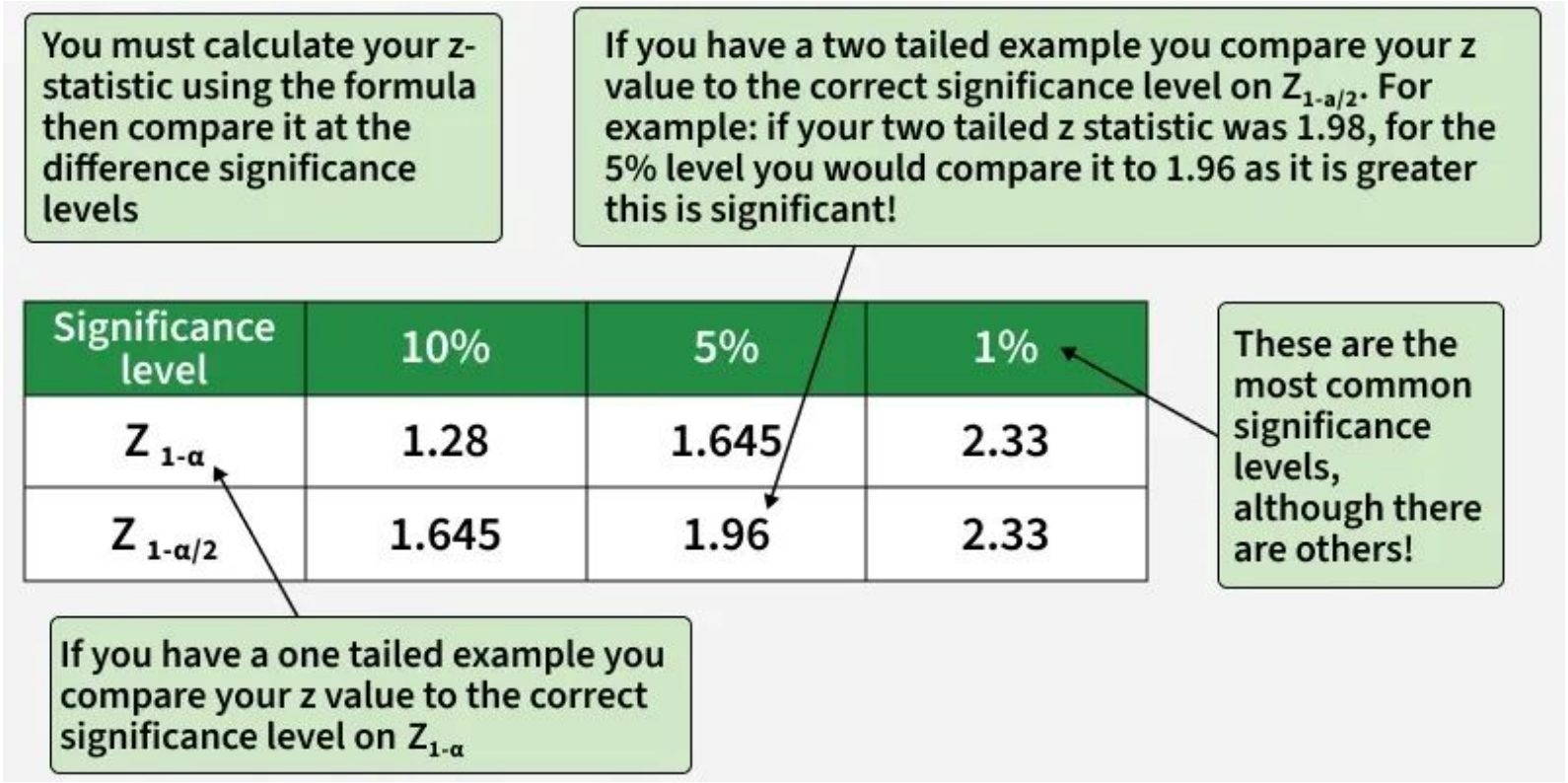
- $\bar{x}$: mean of the sample.
- μ : mean of the population.
- σ : Standard deviation of the population.
- n : Sample size

Z-Test

For a z-test to provide reliable results these assumptions must be met:

1. **Normal Distribution:** The population from which the sample is drawn should be approximately normally distributed.
2. **Equal Variance:** The samples being compared should have the same variance.
3. **Independence**: All data points should be independent of one another.

The Z-Table



Significance level	10%	5%	1%
$Z_{1-\alpha}$	1.28	1.645	2.33
$Z_{1-\alpha/2}$	1.645	1.96	2.33

Example 1

Problem 1: A company claims that the average battery life of their new smartphone is 12 hours. A consumer group tests 100 phones and finds the average battery life to be 11.8 hours with a population standard deviation of 0.5 hours. At a 5% significance level, is there evidence to refute the company's claim?

Solution

Solution:

Step 1: State the hypotheses

$H_0 : \mu = 12 \quad (\text{null hypothesis})$

$H_1 : \mu \neq 12 \quad (\text{alternative hypothesis})$

Step 2: Calculate the Z-score

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$
$$= \frac{11.8 - 12}{\frac{0.5}{\sqrt{100}}}$$
$$= \frac{-0.2}{0.05}$$
$$= -4$$

Step 3: Find the critical value (two-tailed test at 5% significance)

$Z_{0.025} = \pm 1.96$

Step 4: Compare Z-score with critical value
|-4| > 1.96, so we reject the null hypothesis.

Conclusion: There is sufficient evidence to refute the company's claim about battery life.

Step 5: Make a Decision

We compare the test statistic to a critical value from a statistical table or use the p-value:

1. Using Critical Value:

- If test statistic > critical value → reject H_0.
- If test statistic ≤ critical value → fail to reject H_0.

2. Using P-value:

- If p-value ≤ α → reject H_0.
- If p-value > α → insufficient evidence to reject H_0, not proof that H_0 is true.

Example: If p-value is 0.03 and α is 0.05, we reject the null hypothesis because 0.03 < 0.05.

Step 6: Interpret the Results

Based on the decision, we conclude whether there is enough evidence to support the alternative hypothesis or if we fail to reject the null hypothesis.

Two-tailed Test Example

Blood glucose level for obese patients have a mean of 100 and a standard deviation of 15. A researcher thinks that a diet high in raw cornstarch will have a positive or negative effect on blood glucose levels. A sample of 30 patients who have tried the raw cornstarch have a mean glucose level of 140. Test the hypothesis that **the raw cornstarch had an effect**.

One-Tailed Test Example

A principal of a certain school claims that **the students of his school are above the average intelligence**. A random sample of 30 students' IQ scores have a mean score of 112. Is there sufficient evidence to support the claim? The mean population IQ is 100 with a standard deviation of 15.

Types of Z-test

1. One Sample Z test

A one-sample Z-test is used to determine if the mean of a single sample is significantly different from a known population mean. Use it when:

- The population standard deviation is known.
- The sample size is large (usually $n>30$).
- The data is approximately normally distributed.

2. Two-sampled z-test

In this test we have provided 2 normally distributed and independent populations and we have drawn samples at random from both populations. Here we consider u_1 and u_2 to be the population mean and X_1 and X_2 to be the observed sample mean. Here our null hypothesis could be like this:

- $H_0 : \mu_1 - \mu_2 = 0$ and alternative hypothesis
- $H_1 : \mu_1 - \mu_2 \neq 0$

and the formula for calculating the z-test score:

$$Z = \frac{\left(\overline{X_1} - \overline{X_2}\right) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where σ_1 and σ_2 are the standard deviation and n_1 and n_2 are the sample size of population corresponding to u_1 and u_2 . Let's look at the example to understand:

Thank You

Any questions?